

ZIYAN HUANG

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RESEARCH INTERESTS

I am a M.S. Candidate in Mathematics at CSU, working on computational imaging, focusing on:

- **Diffractive imaging and computer vision**, optimizing lightweight, camera-specific back-ends for each individual camera; constructing instance-specific training datasets via experimental design.
- **Uncertainty quantification**, Analyzing and exploiting uncertainty in computational imaging, and leveraging it to guide better end-to-end optical-algorithm design or image reconstruction.
- **Imaging inverse problem**, providing advanced self-supervised, supervised learning or test-time training frameworks, and exploring physics-informed embedding strategies for image restoration

I aspire to bridge my research with real-world applications and collaborate closely with research teams across mathematics, optics, and statistics.

EDUCATION

- **Central South University** 2024.8 - present
M.S. in Mathematics; GPA: 3.61/4.0
Advisor: Assoc. Prof. Hongqiao Wang
Changsha, China
- **Hohai University** 2020.8 - 2024.6
B.S. in Information and Computing Science; GPA: 4.29/5.0, 85.64/100
Nanjing, China

RESEARCH EXPERIENCE

- ◆ **Physen-Noise2Noise: Physics-Guided Self-Supervised Defocus Deblurring with Bias Correction under Low-Light Conditions** 2025-2026
Advisor: Assoc. Prof. Hongqiao Wang (CSU) and Assoc. Prof. Dongliang Tang (HNU)
 - This study presents a self-supervised deep learning framework designed to solve inverse problems under **biased complex noise conditions**. The framework first maps noisy data to a clean image combined with the noise expectation. It then incorporates a physical model into the loss function as an **auto-regressive training strategy**, enabling joint optimization of both the clean image and the noise distribution expectations.
 - A theoretical framework is proposed to elucidate the proposed PN2N method based on joint optimization through natural constraints of physical models, revealing the characteristic natural constraint contained within the PSF.
 - **Publication:** Physen-Noise2Noise: Physics-Guided Self-Supervised Defocus Deblurring with Bias Correction under Low-Light Conditions (**First Author**), received a **major revision** decision from IEEE Transactions on Computational Imaging.
- ◆ **PhyPrune-TTT: Uncertainty-Guided Agent Pruning with Physics-Consistent Test-Time Training for Lightweight Diffractive Computational Imaging** 2026
Advisor: Assoc. Prof. Hongqiao Wang (CSU) and Prof. Bin Fan (Institute of Optics Electronics, CAS)
 - Built a **simulation pipeline** for severely degraded diffractive computational imaging that integrates monolithic front-end point spread functions (PSFs) with coordinate-dependent degradation, ISO modeling, and RGB ISP simulation, enabling supervised training of lightweight reconstruction backends.
 - Proposed an **uncertainty-aware pruning agent** for structured channel pruning of a U-shaped reconstruction network, jointly optimizing image reconstruction quality and computational cost while using MC dropout-based **mask uncertainty** to guide adaptive sparsity regularization.
 - We introduce a lightweight backend with a **test-time training** (TTT) paradigm, where a learned spatially varying PSF surrogate provides **physics-consistent** self-supervision. This improves robustness beyond dataset-level performance and targets practical real-camera use. We **achieves 600ms** for high-quality 4000 $\times$ 6000 TTT.
 - **Publication:** PhyPrune-TTT: Uncertainty-Guided Agent Pruning with Physics-Consistent Test-Time Training for Lightweight Diffractive Computational Imaging (**First Author**). Manuscript in preparation, targeting IEEE Transactions on Pattern Analysis and Machine Intelligence.

◆ UQ-AdaptTTT: Uncertainty-Guided Adaptive Test-Time Training for Physics-Driven Computational Imaging

2026

Advisor: Assoc. Prof. Hongqiao Wang (CSU)

- Propose UQ-AdaptTTT, which leverages per-pixel predictive uncertainty from a **bayesian neural network** to automatically adjust main loss and regularization weights during **test-time training** for physics-consistent adaptive reconstruction.
- Feature a **plug-and-play** design that seamlessly integrates with diverse computational imaging tasks, enhancing reconstruction quality while delivering trustworthy **uncertainty quantification**.
- **Publication:** UQ-AdaptTTT: Uncertainty-Guided Adaptive Test-Time Training for Physics-Driven Computational Imaging (**Corresponding Author: proposed the idea, mentored experiments and implementation.**). In preparation for ICLR 2027.

WORK EXPERIENCE

- **Peking University Changsha Institute for Computing and Digital Economy** 2024.12 - 2025.3
Worked with Prof. Yinyin Zang Research Intern
I developed data pipelines, dialogue classification, and reinforcement learning-based empathy optimization, building datasets, a multi-task classification model, and a multi-dimensional reward function to enhance LLMs' empathetic understanding and response capabilities.
- **Institute of Applied Physics and Computational Mathematics, Beijing** 2024.6 - 2025.6
Worked with Assoc. Prof. Hongqiao Wang Key Member
Constructing high-precision surrogate models with small samples for detonation problems and cylinder test problems using **Gaussian processes**, leveraging **uncertainty quantification** to achieve inverse parameter calibration tasks via forward physical process surrogate models, with a 50% performance improvement compared to traditional methods.
- **Chinese Flight Test Establishment** 2025.8 - 2026.8
Worked with Assoc. Prof. Liang Yan Key Member
Constructing high-precision **surrogate models** and **experimental design** schemes for aircraft wind tunnel tests, engine tests, and in-flight starting tests under small sample conditions, with MAPE accuracy below 5% and design efficiency exceeding 90%.

SKILLS

- **Deep Learning:** Multi-node, Multi-GPU training, LLMs, Self-supervised learning, Network Pruning.
- **Computational Photograph:** Deblur, Denoise, ISP, Deconvolution, Simulation pipeline.
- **Uncertainty Quantification:** Bayesian neural networks, Gaussian processes, Few-shot surrogate model.
- **Experimental Design:** Bayesian optimization, Active learning, Optimal experimental design.
- **Programming Language:** Python, PyTorch, Tensorflow, Scikit-learn, MATLAB.

AWARDS

- **Hohai University Outstanding Student Scholarship** 2020 - 2024
Recipient of multiple university-level honors including the Academic Excellence, Outstanding Student Leader, Science Technology Innovation, Spiritual Civilization, and Academic Progress Scholarships, and awarded the highest comprehensive honor, **the Outstanding Student Pacesetter**, granted to the most outstanding students in the college for their exceptional contributions to the academic community.
- **Central South University Graduate Scholarships** 2024 - 2026
Awarded to top students based on academic performance and research achievements, in recognition of their outstanding contributions to the academic community.